

OPTIMAL HYPERCONTRACTIVITY AND LOG-SOBOLEV INEQUALITIES ON CYCLIC GROUPS $\mathbb{Z}_{m \cdot 2^k}$

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ABSTRACT. For $1 < p \leq q < \infty$ and $n \in \{3 \cdot 2^k, 2^k\}$ with $k \geq 1$, we prove that the Poisson-like semigroup $(P_t)_{t \in \mathbb{R}_+}$ on $\mathbb{Z}_n$, associated with the word length $\psi_n(k) = \min(k, n - k)$, is hypercontractive from L_p to L_q if and only if $t \geq \frac{1}{2} \log \left(\frac{q-1}{p-1} \right)$. We establish sharp Log-Sobolev inequalities with the optimal constant 2, by performing a KKT analysis, and lifting from the base cases $\mathbb{Z}_6$ and $\mathbb{Z}_4$ via a Cooley–Tukey $n \mapsto 2n$ comparison of Dirichlet forms. The general case for arbitrary n remains open.

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1. INTRODUCTION

The hypercontractivity of the Poisson-like semigroup on the cyclic group $\mathbb{Z}_n$ is a long-standing open problem for all n except $n = 2$ and $n = 4$. More precisely, by equipping $\mathbb{Z}_n$ with the normalized counting measure μ_n , the Poisson-like semigroup $(P_t)_{t \in \mathbb{R}_+}$ is defined as the family of maps $P_t : L_\infty(\mathbb{Z}_n, \mu_n) \rightarrow L_\infty(\mathbb{Z}_n, \mu_n)$, which acts on Fourier series by

$$P_t : \sum_{k=0}^{n-1} a_k \chi_k(x) \mapsto \sum_{k=0}^{n-1} e^{-t\psi_n(k)} a_k \chi_k(x),$$

where $\psi_n(k) = \min(k, n - k)$ is the word-length function on $\mathbb{Z}_n$, and $\chi_k(x) = e^{\frac{2\pi i k x}{n}} \in L_\infty(\mathbb{Z}_n)$. The hypercontractivity problem asks for the optimal time $t_{p,q}$ with $1 < p \leq q < \infty$ such that

$$\|P_t f\|_q \leq \|f\|_p \quad \text{for all } t \geq t_{p,q}.$$

For the case $n = 2$, the optimal time $t_{p,q} = \frac{1}{2} \log \left(\frac{q-1}{p-1} \right)$ follows by applying the classical two-point inequality. That inequality was first proved by Bonami [Bon70], later rediscovered by Gross [Gro75b], and was also used by Beckner [Bec75] to obtain the best constants for the Hausdorff–Young inequality. For $n = 4$, Beckner, Janson, and Jerison [BJJ83] obtained the same optimal time $t_{p,q} = \frac{1}{2} \log \left(\frac{q-1}{p-1} \right)$ by a clever reduction from $\mathbb{Z}_2$ to $\mathbb{Z}_4$, though the approach does not extend to other $\mathbb{Z}_{m \times n}$. For $n = 3$, by Wolff’s reduction [Wol07, Corollary 3.1], the optimal time $t_{2,q}$ of P_t on $\mathbb{Z}_3$ coincides the optimal time

$$t_{2,q} = \frac{1}{2} \log \left(\frac{\frac{2}{3} \left(\frac{1}{3} \right)^{\frac{2}{q}-1} - \frac{1}{3} \left(\frac{2}{3} \right)^{\frac{2}{q}-1}}{\left(\frac{2}{3} \right)^{\frac{2}{q}} - \left(\frac{1}{3} \right)^{\frac{2}{q}}} \right)$$

of the simple semigroup $T_t = e^{-t(\text{id} - \mathbb{E}_\delta)}$ on the weighted two-point space $L_\infty(\{-1, 1\}, \frac{1}{3}\delta_{-1} + \frac{2}{3}\delta_1)$, which was computed by Oleszkiewicz and Łatała [LaO00; Ole03]. For the general optimal time $t_{p,q}$ of $(P_t)_{t \in \mathbb{R}_+}$ on $\mathbb{Z}_3$, only asymptotic information is available; see [Wol07, Theorem 2.1]. For $n = 5$, Andersson [And02] determined $t_{2,q} = \frac{1}{2} \log(q - 1)$ for $q \in 2\mathbb{Z}_+$. For $n \geq 6$, Junge, Palazuelos, Parcet, and Perrin [JPPP17] proved partial results: for even n , $t_{2,q} = \frac{1}{2} \log(q - 1)$ for $q \in 2\mathbb{Z}_+$; for odd n , the same holds when $n \geq q$, via a rather involved combinatorial method. Combining Stein’s interpolation method [Ste56] with Gross’s extrapolation technique [Gro75b; Gro75a; Gro06], they further obtained $t_{p,q} \leq \frac{\log 3}{2} \log \left(\frac{q-1}{p-1} \right)$ in those regimes.

Hypercontractivity is also widely studied for semigroups on manifolds. Among the most classical examples, Weissler [Wei80] proved optimal hypercontractivity for the heat and Poisson semigroups on the circle $\mathbb{S}^1$, while Rothaus [Rot80] independently treated the heat case. For the sphere $\mathbb{S}^n$ ($n \geq 2$), Mueller and Weissler [MW82] established optimal hypercontractivity for the heat semigroup. For Poisson-type semigroups on $\mathbb{S}^n$, see Janson [Jan83], Beckner [Bec92], and Frank–Ivanisvili [FI21].

Now we present our main result for the Poisson-like semigroup $(P_t)_{t \in \mathbb{R}_+}$ on $\mathbb{Z}_n$. Note that the standard argument, based on a simple series expansion of $\|P_t(1 + \epsilon f)\|_q$ and $\|1 + \epsilon f\|_p$ at $\epsilon = 0$, shows the universal lower bound $t_{p,q} \geq \frac{1}{2} \log \left(\frac{q-1}{p-1} \right)$. The theorem below records the optimal times $t_{p,q}$ along a dyadic tower of n .

Theorem 1.1. *For $n = 3 \cdot 2^k$ and $n = 2^k$ with $k \geq 1$, we have*

$$\|P_t f\|_q \leq \|f\|_p \quad \Leftrightarrow \quad t \geq \frac{1}{2} \log \left(\frac{q-1}{p-1} \right)$$

for $1 < p \leq q < \infty$.

A standard route to hypercontractivity proceeds through Log-Sobolev inequalities (LSI) by Gross's celebrated work [Gro75b]: $\|P_t f\|_q \leq \|f\|_p$ holds whenever $t \geq \frac{C}{4} \log \left(\frac{q-1}{p-1} \right)$ if and only if the corresponding LSI holds with constant C . Thus, to prove Theorem 1.1, it suffices to establish the following n -LSI with the optimal constant 2 along the above dyadic tower of n . Denote by A_{ψ_n} the generator of semigroup $(P_t)_{t \in \mathbb{R}_+}$, that is

$$(1.1) \quad A_{\psi_n} : \sum_{k=0}^{n-1} a_k \chi_k(x) \mapsto \sum_{k=0}^{n-1} \psi_n(k) a_k \chi_k(x).$$

The case $n = 4$ in the following theorem is due to the work [BJJ83] and Gross's extrapolation technique [Gro75b].

Theorem 1.2. *For $n = 3 \cdot 2^k$ and $n = 2^k$ with $k \geq 1$, we have the following LSI with the optimal constant 2:*

$$\int_{\mathbb{Z}_n} f^2 \log f^2 d\mu_n - \|f\|_2^2 \log \|f\|_2^2 \leq 2 \langle f, A_{\psi_n} f \rangle_{L_2(\mathbb{Z}_n, \mu_n)}, \quad f \in L_2^+(\mathbb{Z}_n, \mu_n).$$

Our proof of Theorem 1.2 is based on a new induction scheme with three key ingredients:

- (1) Auxiliary weights ϕ_4 on $\mathbb{Z}_4$ and ϕ_6 on $\mathbb{Z}_6$ together with their corresponding LSIs. These LSIs are tighter than those for the word-lengths ψ_n ; this refinement is new even when $n = 4$ and plays a key role in the analysis of LSIs.
- (2) Karush–Kuhn–Tucker (KKT) analysis for the 4- and 6-LSI. We develop an efficient way to handle LSIs on $\mathbb{Z}_n$ via KKT analysis, combined with the aforementioned manipulation of the length functions. The specific structure of ϕ_6 introduces a symmetry in the KKT system that makes the analysis tractable.
- (3) An induction from the n -LSI to the $2n$ -LSI under a crucial compatibility condition. We use a Cooley–Tukey factorization of the $2n$ -point discrete Fourier transform (DFT), which expresses a large DFT as a combination of smaller DFTs and yields a comparison of Dirichlet forms at the scales n and $2n$. The choice of the new weights ϕ_4 and ϕ_6 mentioned in (1) is crucial for the base steps $4 \rightarrow 8$ and $6 \rightarrow 12$.

Finally, we remark that the above ideas are also useful for studying LSIs along other towers of the form $m \cdot n^k$. The KKT analysis and the compatibility condition vary in their technical details, depending on the specific value of n and on the particular choice of weights on $\mathbb{Z}_n$, and we will not carry out this analysis in this paper.

The article is organized as follows. After a brief introduction to the LSI formulation and the n -dimensional KKT framework in Section 2, we analyze the KKT systems associated with the LSIs for $n = 6$ and $n = 4$ in Section 3. Section 4 presents a comparison criterion for a pair of weights that allows us to compare the Dirichlet forms at the scales n and $2n$. We then apply it to the transition $n \rightarrow 2n$ of LSIs and establish Theorem 1.2.

2. FOURIER FORMULATION OF LSI ON $\mathbb{Z}_n$ AND KKT FRAMEWORK

Let F_n denote the $n \times n$ discrete Fourier transform (DFT) matrix

$$F_n = \begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & \omega & \omega^2 & \cdots & \omega^{n-1} \\ 1 & \omega^2 & \omega^4 & \cdots & \omega^{2(n-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \omega^{n-1} & \omega^{2(n-1)} & \cdots & \omega^{(n-1)(n-1)} \end{pmatrix},$$

where $\omega = e^{2\pi i/n}$ and, with 0-indexed rows/columns, $(F_n)_{j,k} = \omega^{jk}$ for $0 \leq j, k \leq n-1$. It is well known that $\frac{1}{\sqrt{n}} F_n$ is unitary (see, e.g., [Dav79, Section 2.5]). For a column vector $x \in \mathbb{C}^n$, we write its DFT as

$$\hat{x} = (\hat{x}_0, \dots, \hat{x}_{n-1})^T = F_n x, \quad \text{where} \quad \hat{x}_k = \sum_{j=0}^{n-1} x_j \omega^{jk}.$$

Given $\lambda \in \mathbb{R}_+^n$, define the entropy functional on the vector λ by

$$H_n[\lambda] := \frac{1}{n} \sum_{k=0}^{n-1} \lambda_k^2 \log(\lambda_k^2) - \frac{\|\lambda\|_2^2}{n} \log \left(\frac{\|\lambda\|_2^2}{n} \right) = \frac{1}{n} \sum_{k=0}^{n-1} \lambda_k^2 \log \left(\frac{n \lambda_k^2}{\|\lambda\|_2^2} \right).$$

The functional $H_n[\lambda]$ is homogeneous of degree 2, i.e., $H_n[\alpha \lambda] = \alpha^2 H_n[\lambda]$ for $\alpha > 0$.